CHANDU PRASAD BHAIRAPU



AI Engineer • Agentic AI for the Energy Sector

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ABOUT ME

AI engineer with an MSc in Artificial Intelligence and 3 years of professional experience. At Kniit I designed and built, as sole developer, an agentic knowledge-extraction system for the German energy sector, working with SAP domain experts and validated against SAP-system data. I work across agentic orchestration, semantic retrieval and cloud architecture, and I measure my systems on accuracy, throughput and cost efficiency.

EXPERIENCE

Associate Generative AI Developer - Agentic Indexing Pipeline Engineer for German Energy Sector

Kniit GmbH

November 2025 – present Muldestausee, Germany

- Sole developer owning the software architecture and delivery of an LLM-based knowledge-extraction pipeline for the German energy sector end-to-end (data ingestion, chunking, prompt engineering, and indexing with agentic, deterministic, and monolithic strategies), reporting directly to the CTO and working with SAP domain experts as key stakeholders.
- Designed agentic orchestration with LangGraph (entity-extraction, relationship-analysis, consolidation, and reflection agents running in parallel) for regulatory and technical documents, raising precision/recall from 33%/45% to 93%/90% against an SAP-system-derived ground truth.
- Engineered a deterministic (code-based) pipeline for a second, high-throughput source stream, cutting a $200,000 projected LLM budget to near-zero recurring cost while pushing extraction accuracy toward 100%.
- Designed and implemented a Model Context Protocol (MCP) query service exposing the knowledge graph to AI agents for millisecond-latency retrieval, integrated into Claude Code with custom skills for AI-assisted engineering.
- Developed Expert Interface, a graph curation web app (React frontend, Python microservices backend) on Azure where domain experts review and approve knowledge-graph changes.
- **Tech stack: Python, LangGraph, LangChain, LiteLLM, MCP, GraphRAG, RAG, Vector Search, Neo4j, MongoDB, Postgres, FastAPI, React, Docker, Kubernetes, Azure**

GenAI Developer (AI Consulting Intern) - Development and Deployment of an Agentic AI Workflow

Covestro Deutschland AG

December 2024 – April 2025 Leverkusen, Germany

- Designed an agentic AI application with LangGraph that orchestrates requests across vector database queries, general inference, and summarization, and deployed it on AWS.
- Built the full stack: React frontend and Python backend with FastAPI, connecting the user interface to the agentic backend.
- Containerized the stack with Docker and deployed it to AWS using CDK (infrastructure as code); version-controlled in GitLab with a CI/CD pipeline.
- **Tech stack: Python, LangGraph, LangChain, FastAPI, React, Docker, AWS (CDK, EC2, Lambda), GitLab CI/CD**

GenAI Developer (Master's Thesis) - Evaluating the Performance of RAG Systems

Covestro Deutschland AG

May 2024 – November 2024 Leverkusen, Germany

- Automated AI output quality assurance: evaluated RAG generation and retrieval with RAGAS and PytrecEval against chemist-provided ground truth.
- Benchmarked multilingual embedding models for semantic retrieval on over 1M data chunks of Covestro's internal chemical data.
- Built event-driven AWS infrastructure with Lambda, SQS, and DynamoDB for PrismaChat RAG evaluation using CDK.
- **Tech stack: Python, LangChain, Hugging Face Transformers, RAGAS, PytrecEval, AWS (Lambda, SQS, DynamoDB, S3, CDK), Azure AI Search, Docker**

Data Scientist (Intern) – ML-Based Prediction of Prepolymer Synthesis Outcomes

Covestro Deutschland AG

November 2023 – April 2024 | Leverkusen, Germany

- Built and compared Random Forest, XGBoost, and LSTM models to predict prepolymer synthesis outcomes (e.g. viscosity) from time-series process data.
- **Tech stack: Python, Pandas, Scikit-Learn, XGBoost, TensorFlow/Keras, Streamlit**

Machine Learning Engineer - Electrical Vehicle Load Forecasting using Time Series Analysis

Cognizant Technology Solutions

September 2021 – March 2022 | Hyderabad, India

- Developed ARIMA, SARIMA, XGBoost regressor, FB Prophet, and LSTM models for EV load forecasting; the LSTM achieved 96% test accuracy on unseen data.

PROJECTS

Project: Meter Read Plausibility Agent | Python | LangGraph | MCP | FastAPI | Docker

- Agentic workflow for implausible smart-meter readings at utilities: rule-based checks screen every interval read at high throughput, and only ambiguous cases go to an LLM agent that retrieves the relevant metering rules by semantic retrieval and proposes a resolution for human approval.
- Governance by design: every agent decision is written to an audit log with its sources, personal data is masked before any LLM call, and a labelled test set checks accuracy, cost, and latency in a CI/CD pipeline.

TECHNICAL SKILLSET

- **Agentic AI:** agentic orchestration (LangGraph, LangChain), multi-agent systems, MCP, LLM APIs (LiteLLM), prompt engineering, AI output evaluation (RAGAS, PytrecEval)
- **Semantic Retrieval:** RAG, GraphRAG, vector search, embedding models (Hugging Face Transformers), Neo4j, vector databases (Qdrant, Pinecone, Azure AI Search)
- **Cloud Architecture:** cloud-native services on AWS (CDK, Lambda, EC2, SQS, DynamoDB, S3, VPC), Azure, Docker, Kubernetes, GitLab CI/CD
- **AI-Assisted Engineering:** Claude Code (custom skills, agentic coding workflows)
- **Programming & Data:** Python, SQL (Postgres), MongoDB, FastAPI, REST APIs, React (beginner)
- **Machine Learning:** Scikit-Learn, PyTorch, TensorFlow, XGBoost, time-series analysis

EDUCATION

Master of Science in Artificial Intelligence — March 2022 – April 2025
Technische Hochschule Würzburg-Schweinfurt, Würzburg, Germany

Bachelor of Technology in Mechanical Engineering, Minor in Computer Science & Engineering — August 2017 – July 2021
G Pulla Reddy Engineering College, Kurnool, India

PUBLICATIONS & CERTIFICATIONS

- **[1]** Navigating the Future: An approach of Autonomous Indoor Vehicles
- Udemy - LLMs Mastery : Transformers & Generative AI
- Udemy - Machine Learning A-Z : AI, Python + ChatGPT
- Udemy - The Complete Neural Networks Boot camp

LANGUAGES

English (Business Fluent) | German (B1) | Hindi (Fluent) | Telugu (Native)